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Problem Set for Tutorial 2

To be discussed on T2, September 6, 2019

1. Each school-week night John can play video games, talk on the phone, or watch a movie. He has a total of five hours to spend on one of these three things.

(a) Using the information, fill out the marginal benefit columns below.

Quantity (hours)	Play Video Games		Talk on Phone		Watch a Movie	
	Total Benefit	Marginal Benefit	Total Benefit	Marginal Benefit	Total Benefit	Marginal Benefit
0	0	-	0	-	0	-
1	25		9		6	
2	27		14		9	
3	28		16		12	
4	28		18		15	
5	28		19		18	

- (b) Suppose John plans to play video games for 2 hours and talk on the phone for the other 3 hours. Given this information, John's total benefit from the plan will be _____.
- (c) Suppose John plays video games for 4 hours. For the last hour, he decides to watch a movie over talking on the phone. His total benefit will _____ because the marginal benefit from watching a movie for 1 hour is _____ the marginal benefit from talking on the phone for 1 hour.
- (d) John's total benefit will be maximum if he plays video games for _____ hour(s), talks on the phone for _____ hour(s), and watches a movie for _____ hour(s).
2. Consider the following demand schedule for bags:

Price	Quantity
\$11	56
\$13	52
\$15	44

- (a) When the price of bags rises from \$13 to \$15, the price elasticity of demand is approximately _____ (calculate using the arc elasticity of demand.)
- (b) When the price of bags increases from \$13 to \$15, the total expenditure will _____ because the price elasticity of demand is _____.
3. Suppose the demand for a good is given by

$$Q_D = 4 - P$$

- (a) Graph in an appropriately labeled diagram the demand curve.
 - (b) Suppose that the price is \$1. The consumer surplus is _____.
 - (c) Suppose that the price increases to \$3. The new consumer surplus is _____.
 - (d) The consumer surplus has reduced because _____.
4. The following graph shows Esther's Best affordable point at A .



The price of seeing a movie falls and Esther's new best affordable point is C .

- (a) Draw a budget line that isolates the substitution effect. Label it B_3 .
- (b) Draw a point to indicate where Esther consumes if she faces the budget line you've drawn. Label this point B .
- (c) For Esther, the substitution effect is the movement from _____ and the income effect is the movement from _____.
- (d) The total effect is the movement from _____. Verify that it is indeed the sum of the income and substitution effects.